JSC «Kazakh-British Technical University»
School of IT and Engineering

APPROVED BY
Dean of SITE
Azamat Imanbayev
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SYLLABUS

Discipline: Programming Principles 2
Number of credits: 4 (2/0/2)
Term: Spring 20__
Instructors' full names: Beisenbek M. Baisakov, Askar K. Akshabayev, Nikita A. Ussyukin, Ayan Aksha, Zhandos O. Zhanabekov, Nurzhigit Tolendi, Tomiris Ismagambetova, Akbota Mautkazy, Askhat Yergaliyev, Arnur Kelgenbayev

Personal Information about the Instructor	Time and place of classes		Contact information
	Lessons	Office Hours	e-mail
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Course duration - 4 credits, 15 weeks (60 class hours)

Course prerequisites - Programming Principles I

Course Description

This course provides a comprehensive introduction to Python programming, version control with Git, game development using Pygame, and fundamental database operations. It is designed for beginners and intermediate learners who want to build a strong foundation in programming, problem-solving, and software development practices. Throughout the course, students progress from basic Python syntax to advanced concepts such as object-oriented programming, file handling, regular expressions, and database integration. The course also includes hands-on projects (TSIS), weekly practical assignments and a final examination.

Course Goals, Learning Outcome(s) and Outline:

The main goals of the course are to:

1. Develop a strong understanding of Python programming fundamentals.
2. Teach students to apply control structures, functions, and object-oriented design principles.
3. Introduce version control using Git to support collaborative software development.
4. Build practical skills in working with files, directories, and DB.
5. Introduce regular expressions for pattern matching and text processing.
6. Provide hands-on experience in game development with Pygame.
7. Teach basic database operations including reading, saving, updating, and deleting records.
8. Prepare students for real-world programming tasks through practical assignments and project defenses (TSIS).
9. Build students' confidence in writing clean, structured, and maintainable code.

Upon successful completion of this course, students will be able to:

Python Fundamentals

- Understand Python syntax, variables, data types, numbers, casting, strings, and operators.
- Write programs using conditional statements and basic algorithms.
- Work with built-in data structures: lists, tuples, sets, dictionaries, and arrays.
- Write user-defined functions, use lambda expressions, and implement classes and inheritance.

Advanced Python

- Apply iterators, generators, Python scope rules, modules, dates, math operations, and JSON handling.
- Use regular expressions to search, match, and process text.
- Use Python’s built-in functions effectively.
- Handle files and directories: reading, writing, deleting, and navigating the filesystem.

Pygame Development

- Create graphical applications and simple games using Pygame.
- Work with images, fonts, input handling, timers, sound effects, and scene logic.
- Develop mini-projects such as **Snake** and **Paint**.

Databases

- Connect Python applications to databases.
- Perform CRUD operations: create, read, update, and delete data.
- Understand SQL basics and implement data persistence in Python projects.

Professional Practices

- Use Git for version control and maintain repositories on GitHub.
- Present and defend technical work through structured TSIS assignments.
- Prepare for a comprehensive final exam covering all course content.

Methodology

The teaching methodology includes:

- **Lectures** to introduce theoretical concepts.
- **Hands-on coding sessions** for practical learning.
- **Weekly assignments** to reinforce skills through independent work.
- **Mini-projects (TSISes)** developed throughout the course (e.g., Pygame games).
- **Coding exercises** using Git and GitHub.
- **In-class problem-solving** and algorithmic thinking tasks.
- **TSIS defenses** where students present their work.
- **Final exam** evaluating conceptual and practical understanding.

Materials

Students will use the following resources throughout the course:

Software & Tools

- Python 3.x
- Git & GitHub
- Pygame library
- PostgreSQL

- Code editor (VS Code is highly recommended)

Online Resources

- W3Schools Python Tutorials: https://www.w3schools.com/python/default.asp
- Official Python Documentation: https://docs.python.org/
- GitHowTo Git online guide: https://githowto.com/
- Git documentation: https://git-scm.com/docs
- Pygame documentation: https://www.pygame.org/docs/
- https://www.geeksforgeeks.org/python/generators-in-python/

Course Materials Provided

- Laboratory works (Contests, programming tasks)
- TSIS assignment sets
- Code examples and templates
- Demo projects (Snake, Paint)

COURSE CALENDAR

Class work		
Week	**Topic**	**SIS, TSIS**
1	**L1. Python fundamentals.** 1. Introduction to Python 2. Python interpreter, scripts vs interactive mode 3. Basic syntax & indentation 4. Comments 5. Variables 6. Primitive data types 7. Numbers & type casting 8. Strings 9. String formatting 10. Booleans 11. Operators 12. Conditions (`if/elif/else`) 13. Git basics (init, clone, commit, push)	**Practice 1.** 1. Basic syntax, variables, strings, conditions exercises 2. Problem set 1
2	**L2. Collections & Loops.** 1. `while` loops 2. `for` loops 3. Lists: indexing, slicing, methods 4. Tuples 5. Sets 6. Dictionaries 7. Iteration over collections	**Practice 2.** 1. Loops + lists/tuples/dicts/sets exercices 2. Problem set 2

3	**L3. Functions & OOP Basics.** 1. Functions (`def`, arguments, return) 2. Default parameters 3. Lambda expressions 4. Introduction to OOP 5. Classes, objects, attributes 6. Inheritance 7. Method overriding	**Practice 3.** 1. Functions + simple class hierarchy exercices 2. Problem set 3
4	**QUIZ1: Contest for L1-L3.**	
5	**L4. Advanced Python Constructs.** 1. Iterators 2. Generators (`yield`) 3. Variable scope (local/global/nonlocal) 4. Modules & packages (`import`) 5. Working with dates (`datetime`) 6. Math operations (`math`, `random`) 7. JSON: parsing & serialization	**Practice 4.** 1. Iterators/generators + JSON mini-task exercices 2. Problem set 4
6	**L5. Regular Expressions.** 1. Introduction to Regex 2. Matching, searching, replacing 3. Metacharacters 4. Quantifiers 5. Special sequences 6. Character classes 7. `re.compile()` and efficient pattern usage	**Practice 5.** 1. Regex filters, validation tasks exercices 2. Problem set 5
7	**L6. Built-ins & work with Files.** 1. Built-in Python functions (full overview) 2. Useful builtins: `map`, `filter`, `enumerate`, `zip`, etc. 3. File handling (read/write/append) 4. Working with directories (`os`, `pathlib`) 5. Deleting and creating files	**Practice 6.** 1. File parser + directory operations exercices 2. Problem set 6
8	**QUIZ2: Contest for L1-L6.**	
9	**L7. Databases I (PostgreSQL).** 1. Connect To PostgreSQL Database 2. Create Tables 3. Insert Data Into Table 4. Update Data 5. Query Data 6. Delete Data from Tables	**Practice 7.** 1. Database CRUD basics exercices
10	**L8. Databases II (Transactions & Procedures).** 1. Handle Transactions 2. Call PostgreSQL Functions 3. Call PostgreSQL Stored Procedures 4. Work with BLOB Data	**Practice 8.** 1. DB transactions + stored procedures/funcs exercices 2. TSIS1 - Phonebook

		app
11	**L9. Pygame I.** 1. Getting Started 2. Working with Images 3. Music and Sound Effects 4. Geometric Drawing 5. Timer 6. Fonts and Text 7. More on Input 8. Centralized Scene Logic 9. Paint app demo	**Practice 9.** 1. Basic drawing application exercices 2. TSIS2 - Paint app
12	**QUIZ3: Test quiz for all topics: L1-L9**	
13	**L10. Pygame II.** 1. Sprites 2. Collision detection 3. Movement logic 4. Racer game demo	**Practice 10.** 1. Sprites exercices 2. TSIS3 - Racer game
14	**L11. Pygame III.** 1. Snake game architecture 2. Grid logic 3. Game states 4. Score manager 5. Snake game demo	**Practice 11.** 1. Build core game loop exercices 2. TSIS4 - Snake game
15	**TSIS 1-4 Defense** 1. Final paint, snake and racer game projects 2. Phonebook with PostgreSQL	

COURSE ASSESSMENT PARAMETERS

Type of activity	Final scores
Practice defense and github submission before deadline	1 point per defense (total 11 points)
Practice contest	1 point per contest (total 6 points)
Quiz Contest 1	10 points
Quiz Contest 2	10 points
Quiz Test	10 points
TSIS Defense	13 points
Final exam	40 points
Total	**100 points**

Criteria for evaluation of students during semester:

	Assessment criteria	Weeks																Total scores
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16+	
1.	Github submission + defense		*	*		*	*	*		*	*	*		*	*	*		11%
2.	Lab Contest		*	*		*	*	*		*								6%
3.	Quiz Contest 1				*													10%
4.	Quiz Contest 2								*									10%
5.	Quiz Test												*					10%
5.	TSIS Defense															*		13%
6.	Final exam																*	40%
	Total																	**100%**

ATTENTION!

1) If a student missed more than 30% of lessons without a plausible reason, the said student receives an «F» (Fail) grade;
2) If for two attestations student receives 29 or less points, this student is not accepted to the final exam and for all course he (she) receives «F» (Fail) grade;
3) If student receives on final exam 9.4 or less points, then independently on how many points he (she) received for two attestations, in whole he (she) receives «F» (Fail) grade;
In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives «F» (Fail) grade.
4) In case of non-compliance of the course project with the given assignment, the student is not allowed to defend the course project and automatically receives an «F» (Fail) grade.
5) Practical work must be uploaded to the Team's assignment in written form. If a student fails to upload the work before the specified deadline, the work will not be considered. A score of zero is given.
6) All written work must be completed according to the standard of KBTU. The standard and template are attached to the TEAMS platform. The Teams team has added a video lesson on preparing reports according to standards.
7) All text materials are available on the https://wsp.kbtu.kz/ and MS Teams platforms.
8) If a student does not pass the practice task on time, then admission to the next practice will be only after the mandatory completion of the previous work. Points for previous work will be lost, but feedback will be given on correct completion.
9) To qualify for the exam, each student must individually pass and complete 10 practical works.
10) If plagiarism is detected in the text of the work, in the program code, or if the work does not meet the standards, the work is canceled.

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute “participation”. Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).

4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.
6. Make-up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

Grading Rubric

Next rubric defines how students' work will be evaluated across practices, quizzes, contests, and the TSIS defense.

A. Practice Assignment Defenses (11 points total)

Each practice session assesses correctness, completeness, clarity of explanation, and code quality.

Criterion	Excellent (1.0 / 1.0)	Satisfactory (0.5 / 1.0)	Unsatisfactory (0.0 / 1.0)
Correctness	Solution works fully and meets requirements	Solution partially works	Solution does not work
Completeness	All tasks completed	Majority completed	Incomplete
GitHub submission	Submission is before the deadline	—	Submission is after the deadline
Explanation / Defense	Clear reasoning, confident answers	Basic understanding	Cannot explain solution

Formula for calculating points for the defense: AVG = SUM(Criterion) / 4

B. Lab Contests (6 points total)

Maximum grade is earned with 100% correctly solved problems. Partial solutions are not accepted. The final grade directly corresponds to the percentage of problems solved.

C. Contest Quizzes (20 points total)

Each contest is graded based on passed tests. Partial solutions are allowed.

- Q1 Contest (10 pts)

- Q2 Contest (10 pts)

Contests are graded based on the percentage of correctly solved problems. 100% of solved problems will earn the maximum grade for the contest, otherwise the grade will be adjusted according to the percentage.

Formula for contest quiz grade: Grade = (student_score / max_possible_score) * 10

D. Test Quiz (10 points total)

Grading for the test quiz uses a standard percentage-based system. Maximum grade is earned with 100% correct answers and the final grade directly corresponds to the percentage of correct responses. Test's questions can be:

- open questions
- multiple choice questions with one answer
- multiple choice questions with multiple answers
- code-based questions

E. TSIS Defense (13 points)

Criterion	Excellent (3)	Good (2)	Fair (1)	Poor (0)
Functionality	Project works flawlessly	Minor bugs	Works partially	Does not run
Complexity	Above requirements	Meets requirements	Basic implementation	Below expectations
Code Quality	Structured, modular, readable	Some structure	Minimal structure	Disorganized code
Presentation	Clear, professional, confident	Good explanation	Basic explanation	Cannot explain code

Formula for calculating points for the TSIS defense: AVG = SUM(Criterion) / 12 * 13

Rubric for the overall course grade:

Grade	Achievement percentage	Assessment criterion

«Excellent»	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
	90-94%	
«Good»	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks; Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
	80-84%	
	75-79%	

	70-74%	
«Satisfactory»	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.
	60-64%	
	55-59%	
	50-54%	

Minutes #1 of the School of Information Technology and Engineering meeting on August 18, 2025